

The later C^* -uniqueness answer to arXiv:2107.05806

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered` (Question 9.2: necessary and sufficient conditions for uniqueness of a C^* -norm).

The exact source question

Mori's paper introduces R^* -algebras—algebraic pre- C^* -algebras, or equivalently regular $*$ -subalgebras of bounded operators—and proves that every R^* -algebra has a unique C^* -norm. In its Questions section it asks:

Does the study of R^* -algebras give a better understanding of the uniqueness of C^* -norm?
In particular, can we obtain a nice necessary/sufficient condition for a $*$ -algebra to have a unique C^* -norm?

This is Question 9.2 of the published numbering [1].

The later affirmative answer

An and Gao explicitly reproduce this as their Question 2 and state that Section 4 answers it affirmatively [2]. Their results give not merely one criterion but several equivalent descriptions for a pre- C^* -algebra $\mathcal{A}$:

- Theorem 4.1 characterizes C^* -uniqueness through the $*$ -representations of $\mathcal{A}$.
- Theorem 4.4 gives an equivalent formulation through ideals of the enveloping C^* -algebra $C^*(\mathcal{A})$.
- Theorem 4.6 reformulates it in the primitive ideal space of $C^*(\mathcal{A})$, and Theorem 5.6 gives a weak-containment criterion for representations.

There is also a concrete spectral test. Theorem 4.7 says that a unital pre- C^* -algebra $\mathcal{A}$ has a unique C^* -norm if and only if, for every pair of injective $*$ -representations

$$\pi_i : \mathcal{A} \longrightarrow B(H_i) \quad (i = 1, 2)$$

and every self-adjoint $a \in \mathcal{A}$, the operator $\pi_1(a)$ is invertible exactly when $\pi_2(a)$ is invertible. Thus the invertibility of self-adjoint elements is representation-independent. Theorem 4.10 provides the corresponding R^* -algebra formulation, recovering uniqueness of the C^* -norm for every R^* -algebra.

Conclusion and scope

This is an exact literature resolution of the concrete existence question: the later paper names Mori's question, answers it affirmatively, and supplies multiple necessary-and-sufficient conditions. It does not settle the other independent questions in Mori's Section 9 (for example the Kadison finite generation problem, tensor products of hypothetical non-ultramatrixial R^* -algebras, or the noncommutative Kalton–Roberts problem). The result classification here is therefore deliberately restricted to Question 9.2.

References

- [1] M. Mori, *On regular $*$ -algebras of bounded linear operators: A new approach towards a theory of noncommutative Boolean algebras*, arXiv:2107.05806, <https://arxiv.org/abs/2107.05806>.
- [2] G. An and M. Gao, *R^* -algebras and C^* -uniqueness of pre- C^* -algebras*, Journal of Mathematical Analysis and Applications 526 (2023), 127341, <https://doi.org/10.1016/j.jmaa.2023.127341>.